

# Joshua C. Macdonald

Computational Scientist

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## Research Profile

Computational scientist whose work determines what actions to take, what experiments to run, and what measurements are worth collecting in systems where interventions are costly and uncertainty is unavoidable. I develop methods, scientific AI/ML, and research software for partially observed systems across earth, environmental, and health sciences — integrating generative modeling (mechanistic, statistical, and hybrid), Bayesian inference, uncertainty quantification, and scientific computing to build defensible decision support. Current application areas include infectious disease forecasting and intervention timing, surveillance design, marine and terrestrial ecology, and cultural transmission dynamics.

## Education

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|------------|------------------------------------------------------------------------------------------|---------------------------------|
| <b>PhD</b> | <b>University of Louisiana at Lafayette</b> , Mathematics                                | Lafayette, Louisiana, USA       |
|            | • Advisor: Hayriye Gulbudak                                                              | 2017 – 2022                     |
|            | • Dissertation: <i>Population Dynamics Under Disturbance in Ecology and Epidemiology</i> |                                 |
| <b>MS</b>  | <b>University of Louisiana at Lafayette</b> , Mathematics                                | Lafayette, Louisiana, USA       |
|            |                                                                                          | 2017 – 2020                     |
| <b>BA</b>  | <b>University of North Carolina Asheville</b> , Mathematics                              | Asheville, North Carolina, USA  |
|            |                                                                                          | 2014 – 2016                     |
| <b>BA</b>  | <b>University of North Carolina Greensboro</b> , Archaeology                             | Greensboro, North Carolina, USA |
|            |                                                                                          | 2008 – 2013                     |

## Appointments

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| <b>Johns Hopkins Bloomberg School of Public Health</b> , Assistant Scientist                                                                                                                                                                                                             | Baltimore, Maryland, USA |
| International Vaccine Access Center, Department of International Health (Global Disease Epidemiology and Control). Mentor: Shaun Truelove. Research focus: infectious disease modeling and inference, scientific computing, decision support systems, and research software engineering. | 2026 – present           |
| <b>Johns Hopkins Bloomberg School of Public Health</b> , Postdoctoral Scholar                                                                                                                                                                                                            | Baltimore, Maryland, USA |
| International Vaccine Access Center, Department of International Health (Global Disease Epidemiology and Control). Mentor: Shaun Truelove. Research focus: infectious disease modeling and inference, scientific computing, decision support systems, and research software engineering. | 2024 – 2026              |
| <b>Tel Aviv University</b> , Zuckerman STEM Leadership Postdoctoral Fellow                                                                                                                                                                                                               | Tel Aviv, Israel         |
| School of Zoology. Mentor: Yoav Ram. Research drew directly on archaeological training in the study of material culture diversity and cultural transmission, applying Bayesian dimensionality reduction and multivariate data analysis to cultural datasets.                             | 2022 – 2024              |

## Computational Expertise

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**Modeling:** Generative models (ODE/PDE/stochastic/hybrid); compartmental and agent-based models; scientific AI/ML (physics-embedded surrogates, lawful learning); multivariate data analysis and dimensionality reduction; stability, bifurcation, and sensitivity analysis; parameter identifiability and inverse problems

**Inference:** Bayesian hierarchical models; variational and simulation-based inference; data assimilation; uncertainty quantification; model calibration and validation; proper scoring rules and forecast evaluation

**Computing:** Python, Julia, C++, Stan, R, MATLAB. Scientific computing; high-performance numerical solvers; operator splitting; data pipelines and exploratory analysis; reproducible workflows (Git, CI/CD, containerization)

## Published Papers

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**Within-host viral growth and immune response rates predict FMDV transmission dynamics for African buffalo** 2024

*J.C. Macdonald*, H. Gulbudak, B. Beechler, E. Gorsich, S. Gubbins, E. Perez, A. Jolles

[10.1086/730703](https://doi.org/10.1086/730703) (The American Naturalist)

**Cultural transmission, networks, and clusters among Austronesian-speaking peoples** 2024

*J.C. Macdonald*, J. Blanco-Portillo, M.W. Feldman, Y. Ram

[10.1017/ehs.2024.45](https://doi.org/10.1017/ehs.2024.45) (Evolutionary Human Sciences)

**Forward hysteresis and Hopf bifurcation in an NPZD model with application to harmful algal blooms** 2023

*J.C. Macdonald*, H. Gulbudak

[10.1007/s00285-023-01969-7](https://doi.org/10.1007/s00285-023-01969-7) (Journal of Mathematical Biology)

**First and second wave outbreak size with reactive quarantine measures: a COVID-19 model applied to an epidemic in China** 2022

C.J. Browne, H. Gulbudak, J.C. Macdonald

[10.1016/j.jtbi.2021.110919](https://doi.org/10.1016/j.jtbi.2021.110919) (Journal of Theoretical Biology)

**Modeling COVID-19 outbreaks in the United States with distinct testing, lockdown speed, and fatigue rates** 2021

*J.C. Macdonald*, C.J. Browne, H. Gulbudak

[10.1098/rsos.210227](https://doi.org/10.1098/rsos.210227) (Royal Society Open Science)

**An immuno-epidemiological model of foot-and-mouth disease in African buffalo with carrier transmission** 2026

S. Atkins, H. Gulbudak, S. Welker, H. Smith, *J.C. Macdonald*, B.R. Beechler, A.E. Jolles

(Bulletin of Mathematical Biology)

## Papers Under Review

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**Multi-Model Insights into Racial and Ethnic Heterogeneities in Early COVID-19 Transmission and Severity in the U.S.** 2026

S.J. Bents, S.L. Larsen, S. Jung, E. Howerton, et al., including *J.C. Macdonald*

(In revision, American Journal of Epidemiology)

## Preprints

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**Bayesian estimation of the number of significant principal components for cultural data** 2024

*J.C. Macdonald*, J. Blanco-Portillo, M.W. Feldman, Y. Ram

[10.48550/arXiv.2409.12129](https://doi.org/10.48550/arXiv.2409.12129)

## Manuscripts in Preparation

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- Posterior predictive eigenvalue testing for spectral dimensionality selection** 2026  
*J.C. Macdonald*, S. Naim, Y. Ram  
(In preparation for Bioinformatics or PLOS Computational Biology)
- What Should Structural Evaluation Mean?: Label-Free Structural Evaluation Scores for Scientific Prediction** 2026  
*J.C. Macdonald*, M. Marshall, S. Truelove  
(In preparation for Statistical Science)
- Decision Support Modeling for One Health Pathogens: Crimean-Congo Hemorrhagic Fever Surveillance and Control Design** 2026  
*J.C. Macdonald*, M. Kugler, B. Beechler, A. Jolles  
(In preparation for Royal Society Interface)
- Shapelet analysis of infectious disease forecasts reveals heterogeneity across epidemic phases** 2026  
K. Roosa, L.A. White, T.M. Leon, *J.C. Macdonald*, A. Srivastava, S. Truelove  
(In preparation for Epidemics)

## Research Support

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- National One Health Mission India (\$1M / 3 years) — Pending** In review  
Development of an Integrated One Health Disease Modelling and Digital Health Program in Human, Livestock, and Wildlife. Co-Investigator. Research role: mathematical model parameterization and data analysis; mechanistic modeling for HPAI and anthrax in India; stakeholder training in modeling; and supervision of postdoctoral and doctoral trainees.
- USDA-NIFA AFRI US–UK Collaboration (Award #1032897)** 2025 – 2028  
Quiescence to emergence: Does nairovirus diversity loss increase disease transmission, severity and the risk of emergence? US PIs: Brianna Beechler, Anna Jolles, Cameron Browne, and Hayriye Gulbudak; linked UKRI award BB/Z516934/1 with Emma Thomson, Teresa Lambe, Roman Biek, and Georgina Limon-Vega. Collaborator (40% FTE). Research role: mathematical model parameterization, data analysis, and advising on mathematical and statistical methods.
- USDA-ARS Cooperative Agreement (3022-32000-021-018-S)** 2025 – present  
Investigating the Epidemiology of CCHFV in an Endemic Setting: Uganda. PIs: Bree Beechler and Anna Jolles (Oregon State University). Collaborator. Research role: mathematical modeling of cross-species CCHFV transmission dynamics, surveillance design optimization, and advising on mathematical and statistical methods for decision support.
- CDC-RFA-FT-23-0069 (NU38FT000012)** 2023 – present  
Atlantic Coast Center for Infectious Disease Dynamics and Analytics (ACCIDDA). Directors: Justin Lessler, Kimberly Powers, and Shaun Truelove. 60% FTE. Insight Net center for infectious disease forecasting and outbreak analytics. Research role: operational modeling, forecasting pipeline development, and computational infrastructure.
- NSF EEID / UKRI UK–US Collaboration Grant (\$4.05M / 5 years)** 2022 – 2027  
Multi-scale infection dynamics from cells to landscapes: FMD in African buffalo. PIs: Brianna Beechler, Roman Biek, Hayriye Gulbudak, Erin Gorsich, Simon Gubbins, Anna Jolles, and Jan Medlock. Senior Personnel (from 2024). Research role: mathematical model development and parameterization of within-host viral and immune dynamics.

## Zuckerman Postdoctoral Scholarship (\$120K total)

2022 – 2024

Co-evolution of Cultural and Genetic Traits. Principal Investigator. Competitive postdoctoral fellowship; authored and defended research proposal.

## Research Software – Lead Developer

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### OP System (Operator-Partitioned System)

2026 – present

ACCIDDA. Declarative specification language and compiler for structured dynamical systems. Researchers define models via explicit equations or transition diagrams with multi-axis stratification; specifications compile to validated, safe bytecode closures with structured metadata passed to downstream solvers. Specification layer of the operator-partitioned simulation stack.

- [github.com/ACCIDDA/op\\_system](https://github.com/ACCIDDA/op_system)

### pp-eigentest

2026 – present

Posterior predictive eigenvalue testing for determining signal rank in high-dimensional datasets. Three-layer consensus architecture combining dimensionality heuristics, adaptive thresholding, and multiple testing correction (FWER, FDR). NumPy, C++, JAX (GPU), and sparse backends; model-agnostic, consuming posterior summaries from any generative engine (VBPCApY, PPCA, empirical-Bayes Gaussian).

- [github.com/yoavram-lab/pp-eigentest](https://github.com/yoavram-lab/pp-eigentest)

### Trade Study (Python + Julia)

2026 – present

Design and evaluation framework for scientific simulation studies. Simulators with known latent state provide controlled environments to score competing configurations — model formulations, solver choices, measurement strategies, or any design decision — against ground truth using proper scoring rules and multi-objective Pareto optimization. Hierarchical phases, Bayesian model stacking, and global sensitivity analysis. Available in Python and Julia.

- [github.com/jcm-sci/trade-study](https://github.com/jcm-sci/trade-study) (Python)
- [github.com/jcm-sci/TradeStudy.jl](https://github.com/jcm-sci/TradeStudy.jl) (Julia)
- [pypi.org/project/trade-study/](https://pypi.org/project/trade-study/)
- [doi:10.5281/zenodo.19599838](https://doi.org/10.5281/zenodo.19599838)

### OP Engine (Operator-Partitioned Engine)

2025 – present

ACCIDDA. Operator-partitioned ODE/PDE solver core. Splits the right-hand side into explicit and implicit parts, advancing them jointly via nine methods (explicit, IMEX, fully implicit) with adaptive step-size control. Cached implicit solves with dense/sparse autodispatch; zero per-step allocation. Consumes OP System specifications.

- [github.com/ACCIDDA/op\\_engine](https://github.com/ACCIDDA/op_engine)

### vbpcapy

2024 – present

Variational Bayesian PCA for incomplete data with native per-entry missingness handling, full posterior uncertainty quantification, automatic component pruning, and built-in model selection. Missing-aware preprocessing pipeline and C++-accelerated kernels; scikit-learn-compatible API. Applied to genetic, cultural, and ecological datasets.

- [pypi.org/project/vbpcapy/](https://pypi.org/project/vbpcapy/)
- [doi:10.5281/zenodo.19389250](https://doi.org/10.5281/zenodo.19389250)

## Research Software – Collaborative Infrastructure

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### FlePiMoP2 (Flexible Epidemic Modeling Pipeline 2)

2025 – present

ACCIDDA. Contributor to the configuration-driven orchestration engine for CDC-supported infectious disease forecasting and scenario analysis. Plugin architecture decouples model specification, numerical integration, and output persistence.

- [Documentation](#)

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| <b>TabularEpi</b>                                                                                                                                                                                                                                | 2025 – 2025 |
| UNC / ACCIDDA. Tabular epidemic simulator supporting mixed deterministic-stochastic models. Conducted systematic performance benchmarking that identified bottlenecks, informing the design of the next-generation pipeline.                     |             |
| <b>FlepiMoP (Flexible Epidemic Modeling Pipeline)</b>                                                                                                                                                                                            | 2024 – 2025 |
| Hopkins IDD / ACCIDDA. Developed simulation and statistical inference components for operational influenza modeling; conducted benchmarking that identified 5–20x performance improvements, informing the next-generation pipeline architecture. |             |

## Research Software – Operational Deployments

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| <b>Hib Vaccination Modeling (Navajo Nation)</b>                                                                                                                                                                                                                                                                                                            | 2026 – present |
| ACCIDDA. Technical supervisor managing PhD students in implementation of an age- and immune-status-structured Hib model for the Navajo Nation to evaluate the impact of long-running vaccination programs.                                                                                                                                                 |                |
| <b>Flu Scenario Modeling Hub</b>                                                                                                                                                                                                                                                                                                                           | 2024 – present |
| ACCIDDA. Lead Model Developer across the 2024/25, 2025/26, and 2026/27 seasons. Set up the ACCIDDA flu model and operational pipeline (2024/25). Extensive work with external modeling packages drove contributions that led to FlepiMoP2 (2025/26). Supervising an undergraduate student in a scientific overhaul of the flu model for the 2026/27 round. |                |
| <ul style="list-style-type: none"> <li>• <a href="https://github.com/midas-network/flu-scenario-modeling-hub">github.com/midas-network/flu-scenario-modeling-hub</a></li> </ul>                                                                                                                                                                            |                |
| <b>COVID-19 Scenario Modeling Hub</b>                                                                                                                                                                                                                                                                                                                      | 2025 – 2025    |
| ACCIDDA. Support. Data cleaning and assimilation in support of ACCIDDA's health heterogeneities COVID-19 round.                                                                                                                                                                                                                                            |                |
| <ul style="list-style-type: none"> <li>• <a href="https://github.com/midas-network/covid19-scenario-modeling-hub">github.com/midas-network/covid19-scenario-modeling-hub</a></li> </ul>                                                                                                                                                                    |                |

## Invited Talks and Seminars

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| <b>Cross-Scale Feedback Motifs: Structure-Preserving Models and Computational Tools for Complex Systems</b>                                                                        | Mar 2026    |
| Biological Oceanography Seminar, Woods Hole Oceanographic Institution, Woods Hole, Massachusetts                                                                                   |             |
| <b>Decision-Support Modeling for One Health Pathogens</b>                                                                                                                          | Mar 2026    |
| <a href="#">Differential Equations Seminar</a> , University of Maryland Baltimore County, Baltimore, Maryland                                                                      |             |
| <b>Modeling long-term viral infections</b>                                                                                                                                         | 2025        |
| Mathematics Seminar, Shippensburg University, Shippensburg, Pennsylvania                                                                                                           |             |
| <b>Scenario and forecast modeling for respiratory viruses</b>                                                                                                                      | 2025        |
| Navajo Nation Epidemiology Center Data Summit, Window Rock, Arizona                                                                                                                |             |
| <b>Impacts of host behavior and viral life history on infectious disease transmission</b>                                                                                          | 2024        |
| Department of Mathematics, Virginia Tech; Infectious Disease Dynamics Group, Johns Hopkins University                                                                              |             |
| <b>Within-host viral life histories predict FMDV transmission dynamics for African buffalo</b>                                                                                     | 2024        |
| Hawlena Lab, Ben-Gurion University, Midreshet Ben Gurion, Israel                                                                                                                   |             |
| <b>Modeling process-based effects of harmful algal blooms</b>                                                                                                                      | 2023 – 2024 |
| University of Essex (2024); Tel Aviv University (2023); Israel Oceanographic and Limnological Research, Haifa (2023); Inter-University Institute for Marine Sciences, Eilat (2023) |             |

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| <b>Patterns of cultural variation in the Austronesian expansion</b><br>Fortunato Lab, University of Oxford (virtual) | 2023 |
| <b>Consilience in FMDV ecology</b><br>Gresham Lab, New York University; Obolski Lab, Tel Aviv University             | 2022 |

## Selected Conference Presentations

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| <b>Decision-Support Modeling for One Health Pathogens: Using Mechanistic Models for Surveillance and Forecast Design</b><br>Society for Mathematical Biology Annual Meeting, Graz, Austria; mini-symposium talk                                                                                                                         | 2026        |
| <b>Recovering Ecological Geometry: A Trait- and Depth-Structured IPDE Model of Plankton Dynamics</b><br>Society for Mathematical Biology Annual Meeting, Edmonton, Alberta, Canada; mini-symposium talk                                                                                                                                 | 2025        |
| <b>Biologically Grounded Surrogate-Driven Parameter Inference for Sparsely Observed Dynamical Systems</b><br>Symposium on Biomathematics and Ecology Education and Research (BEER), Fairfax, Virginia; contributed talk                                                                                                                 | 2025        |
| <b>Forward hysteresis and Hopf bifurcation in an NPZD model with application to harmful algal blooms</b><br>Society for Mathematical Biology Annual Meeting, Columbus, Ohio; mini-symposium talk                                                                                                                                        | 2023        |
| <b>Within-host viral growth and immune response rates predict FMDV transmission dynamics for African buffalo</b><br>SIAM TXLA, Lafayette, Louisiana (2023); SMB Epidemiology-Immunology Midyear Meeting (2022, virtual)                                                                                                                 | 2022 – 2023 |
| <b>Dengue immunodynamics: antibody-dependent enhancement, cross-reactive immunity, and model-data confrontation</b><br>SIAM TXLA (2023); Mathematical Perspectives on Immuno-Biology, Bulgaria (2023); Center for Combatting Pandemics, Tel Aviv (2023); European Conference on Theoretical and Mathematical Biology, Heidelberg (2022) | 2022 – 2023 |
| <b>Disease dynamics necessarily operate across scales</b><br>SIAM TXLA, South Padre Island, Texas; mini-symposium talk                                                                                                                                                                                                                  | 2021        |
| <b>Modeling COVID-19 outbreaks in the United States with distinct testing, lockdown speed, and fatigue rates</b><br>Society for Mathematical Biology Annual Meeting (virtual); contributed talk                                                                                                                                         | 2021        |

## Workshop Facilitation

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| <b>Insight Net Third Annual Meeting</b> , Tools Workshop Presenter<br>Presenting FlepiMoP2 and the op_system/op_engine simulation stack at the Insight Net Third Annual Meeting tools workshop, Friday Center, University of North Carolina at Chapel Hill.                | Chapel Hill, North Carolina, USA<br>Apr 2026 |
| <b>Insight Net Second Tech Transfer Workshop</b> , Technical Support<br>ACCIDDA. Technical support for the Insight Net Second Tech Transfer Workshop at UNC Chapel Hill. Assisted participants with Python package installation and environment setup for Epydemics tools. | Chapel Hill, North Carolina, USA<br>Jan 2026 |

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| <b>ACCIDDA</b> , Workshop Facilitator                                                                                                           | Baltimore, Maryland, USA |
| Facilitator for the Applied Modeling in Public Health (AMPH) Workshop, October 15–17, 2025,<br>Johns Hopkins Bloomberg School of Public Health. | Oct 2025                 |

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| <b>ACCIDDA</b> , Workshop Facilitator                                                                                                          | Baltimore, Maryland, USA |
| Facilitator for the Applied Modeling in Public Health (AMPH) Workshop, August 12–15, 2024,<br>Johns Hopkins Bloomberg School of Public Health. | Aug 2024                 |

## Teaching

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| <b>Johns Hopkins University</b> , Teaching Assistant                                                                                   | Baltimore, Maryland, USA |
| Teaching assistant for <i>Topics in Infectious Disease Epidemiology</i> with Shaun Truelove and Sara Loo, Summer 2025 and Summer 2026. | 2025 – 2026              |

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| <b>University of Louisiana at Lafayette</b> , Teaching Assistant and Instructor of Record                                                                                                                                                                | Lafayette, Louisiana, USA |
| Instructor of record for elementary statistics (3 sections), survey of calculus (2 sections), precalculus (2 sections), and college algebra (2 sections). Teaching assistant for calculus, linear algebra, differential equations, and basic statistics. | 2017 – 2020               |

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| <b>McDowell County Schools / Memorial University / UNC Asheville</b> , Teaching and Tutoring                                                                                                                    | 2016 – 2017 |
| Instructor of record for North Carolina Math III / Algebra II (2017). Mathematics and statistics tutoring at Memorial University of Newfoundland (2016–2017) and University of North Carolina Asheville (2016). |             |

## Mentoring

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| <b>Mentoring students and postdoctoral researchers</b>                                                                                                                                                                                                                                                                                                                                                           | 2021 – present |
| <ul style="list-style-type: none"> <li>• Jolles and Beechler Labs, Oregon State University (2025–present)</li> <li>• Society for Mathematical Biology (2022–present)</li> <li>• Association for Women in Mathematics (2023–2025)</li> <li>• Ram Lab, Tel Aviv University (2022–2024)</li> <li>• MIDAS Network (2021–2022)</li> <li>• Gulbudak Group, University of Louisiana at Lafayette (2021–2022)</li> </ul> |                |

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| <b>Undergraduate Research Supervision: Shany Naim</b>                                                                                                      | 2024 |
| Tel Aviv University. Supervised undergraduate research project: initial translation of the VBPCAp variational Bayesian PCA codebase from MATLAB to Python. |      |

## Community Outreach

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| <b>Science communication and research dissemination</b>                                                                                                                                                                                                                                                                                     | 2021 – 2022 |
| <ul style="list-style-type: none"> <li>• University of Louisiana R1 Showcase: selected to present COVID-19 modeling research at the university-wide showcase (2022)</li> <li>• Ocean View United Methodist Church, Oak Island, NC: community discussion on vaccines and COVID-19, with on-site pharmacist for vaccination (2021)</li> </ul> |             |

## Professional Service

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| <b>SMB Diversity, Equity, and Inclusion Committee</b> | 2023 – 2025 |
| Member                                                |             |

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| <b>Session Chair / Leader</b>                                                                                                                                                                                                                                                                           | 2021 – 2023 |
| <ul style="list-style-type: none"> <li>• SMB Early Career Workshop: Applying for postdoctoral funding (2023)</li> <li>• SMB Epidemiology-Immunology Midyear Meeting: Contributed talks (2022)</li> <li>• SMB Annual Meeting: CT04-MEPI contributed talks in mathematical epidemiology (2021)</li> </ul> |             |

### Mini-symposium Co-organizer

2019 – 2023

- SIAM TXLA: Data-driven mathematical and statistical modeling approaches to population biology (2023)
- SMB Annual Meeting: Disease Dynamics Across Scales (2023)
- SMB/ESTMB Joint Meeting: Multi-scale and data-driven modeling approaches in ecology, immunology, and epidemiology (2022)
- SIAM TXLA: Reproducibility, reliability, and robustness: confronting models from across mathematical biology with data (2021)
- SIAM TXLA: Population Dynamics in Ecology and Epidemiology (2019)

### Secretary, University of Louisiana AMS Graduate Student Chapter

2021 – 2022

## Journal Reviewing

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**2022–present:** *PLOS ONE*, *Journal of Biological Systems*, *Journal of Biological Dynamics*, *Frontiers in Public Health*, *Frontiers in Veterinary Science*, *Computational and Applied Mathematics*, *Mathematical Biosciences*, *Scientific Reports*, *PLOS Computational Biology*, and *BMC Infectious Diseases*.

## Honors and Awards

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### Travel Awards

2019 – 2022

SIAM, Society for Mathematical Biology, and University of Louisiana Graduate Research Association

### Rhodes Outstanding Graduate Assistant Award

2021

Department of Mathematics, University of Louisiana at Lafayette. Awarded to two graduate students annually (approximately 50 graduate students total).

### Outstanding Contributed Talk Award (Mathematical Epidemiology)

2021

Society for Mathematical Biology Annual Meeting

### Three Minute Thesis Competition, Finalist

2019

Louisiana Sea Grant

## Professional Affiliations

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Society for Mathematical Biology; Society for Industrial and Applied Mathematics; International Society for Bayesian Analysis; United States Research Software Engineer Association; MIDAS Network; Association for Women in Mathematics; Institute of Industrial and Systems Engineers; Society for Health Systems (SHS/IISE); and Healthcare Information and Management Systems Society (HIMSS).